

MICROCANONICAL ENSEMBLE

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SUMMARY OF LAST LECTURE

- **Microstate:** specific configuration of a system at the microscopic level (== what each particle is doing: wavefunction in QM, known position and momentum particle-by-particle in classical physics)
- **Macrostate:** macroscopic description of a system (e.g.: values for energy, temperature, volume, pressure, ...)
- **Microcanonical ensemble:** **isolated** system with **$N \gg 1$** elements/ particles (ensemble) characterized **by fixed energy**, with all **microstates equally likely**, $p(n) = 1/\Omega(E)$, with $\Omega(E)$ the number of microstates with energy E
- **Entropy:** (for now) measure of number of possible microstates in a system, $S(E) = k_B \log \Omega(E)$
- In isolated systems, entropy adds (properties of log)
- **Second law of thermodynamics:** removing constraints from a system (== bringing together previously isolated systems), $\Omega \geq \Omega_1 \Omega_2$, $S \geq S_1 + S_2 \rightarrow \delta S > 0$

CONTENTS

- Temperature
- Heat capacity
- Two state system & the Schottky anomaly
- Pressure & volume
- First law of thermodynamics

TEMPERATURE

$$S(E) = k_B \log \Omega(E)$$



$\Omega(E)$, number of microstates with energy E , depends on the energy $\rightarrow$ relation between energy and entropy?

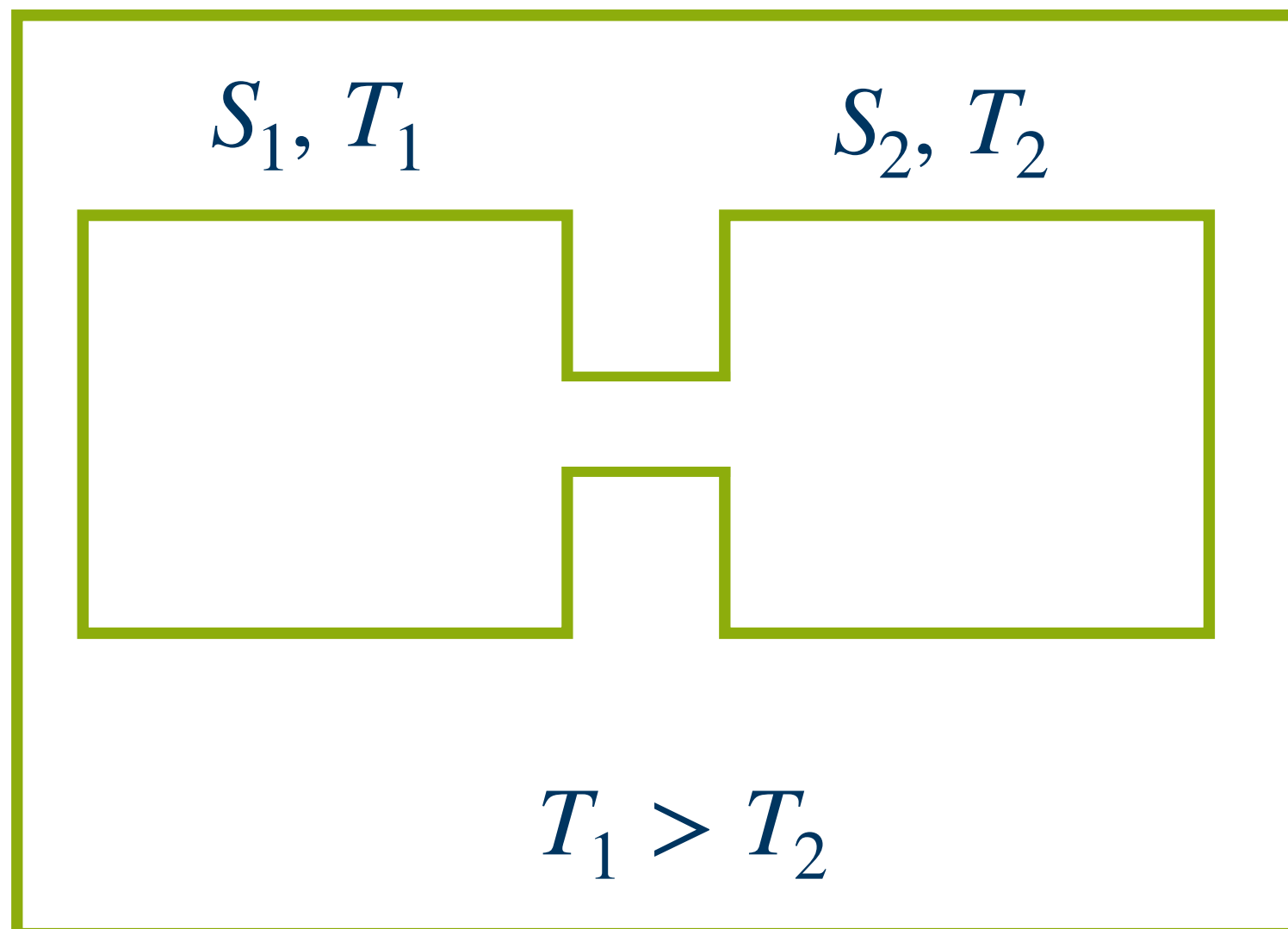
$$\Delta E = \frac{\partial E}{\partial S} \Delta S$$



$$\Delta S = \frac{1}{\frac{\partial E}{\partial S}} \Delta E = \frac{1}{T} \Delta E$$

We have come up with a 'out-of-the-blue' definition for T : how does this definition of T compare with the temperature we are used to?

Isolated



$$\delta S = \frac{\partial S_1(E_1)}{\partial E_1} \delta E_1 + \frac{\partial S_2(E_2)}{\partial E_2} \delta E_2 =$$

$$\left(\frac{\partial S_1(E_1)}{\partial E_1} - \frac{\partial S_2(E_2)}{\partial E_2} \right) \delta E_1 =$$

$$\left(\frac{1}{T_1} - \frac{1}{T_2} \right) \delta E_1 > 0$$

The joint system is isolated: $\delta E_1 + \delta E_2 = 0$

Definition of T

II law of thermodynamics

Q: let's assume $T_1 > T_2$: in which direction does energy flow (== what is the sign of δE_1 ?) Does this match with what we expect from 'our' T ?

Important: this will work with multiple definitions of T ; we will return to why this particular one later (ideal gases)

HEAT CAPACITY: A LINK TO THE MICROSCOPIC WORLD

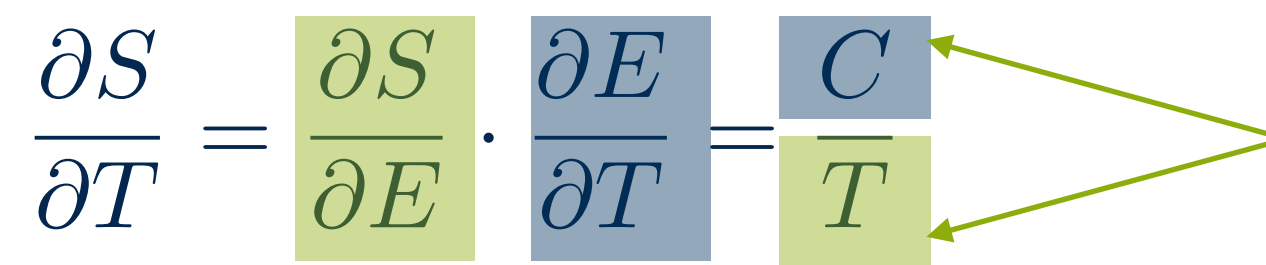
Definition of heat capacity:

$$C = \frac{\partial E}{\partial T}$$

(if you wonder where the “heat” is here, wait for the end of the class)

C is a *macroscopic quantity*: if you add a known ΔE to a system and measure the ΔT , you get $1/C$

but: link to the microscopic world through T

$$\frac{\partial S}{\partial T} = \frac{\partial S}{\partial E} \cdot \frac{\partial E}{\partial T} = \frac{C}{T}$$


We can measure both!!!

Q: look at the definition of heat capacity: do you expect it to have positive or negative sign?

$$\Delta S = \int_{T_1}^{T_2} \frac{C(T)}{T} dT$$

And now we can measure ΔS

THERMODYNAMICALLY STABLE SYSTEM/ 1

Definition of thermodynamically stable system (for now): the system will not undergo spontaneous changes to move to a different state.

Back to last week's case: isolated system put in contact



Number of available states of the combined system:

$$\Omega(E_{total}) = \sum_{\{E_i\}} \Omega_1(E_i) \Omega_2(E_{total} - E_i) = \sum_{\{E_i\}} \exp \left(\frac{S_1(E_i)}{k_B} + \frac{S_2(E_{total} - E_i)}{k_B} \right)$$

$$E_{total} = E_1 + E_2$$

S scales as N, with $N \gg 1 \rightarrow$ we have a $\sum$ of $\exp(N) \rightarrow$ the sum is dominated by its maximum value, where $E_i = E^*$

$$\Omega(E_{total}) \sim \exp \left(\frac{S_1(E^*)}{k_B} + \frac{S_2(E_{total} - E^*)}{k_B} \right)$$

E^* is the energy that maximises the argument of the exp

THERMODYNAMICALLY STABLE SYSTEM/ 2

$$\Omega(E_{total}) \sim \exp \left(\frac{S_1(E^*)}{k_B} + \frac{S_2(E_{total} - E^*)}{k_B} \right)$$

A thermodynamically stable system has reached maximum entropy: we want the maximum of $\Omega(E_{total})$

To find the maximum:

$$\frac{\partial S_1(E^*)}{\partial E} + \frac{\partial S_2(E_{total} - E^*)}{\partial E} = 0 \quad \leftarrow \text{But is this a maximum or a minimum?}$$

Maximum:

$$\frac{\partial^2 S_1(E^*)}{\partial E^2} + \frac{\partial^2 S_2(E_{total} - E^*)}{\partial E^2} < 0$$

Q: write $\frac{\partial^2 S(E)}{\partial E^2}$ as a function of C : do we need $C > 0$ or $C < 0$, for a thermodynamically stable system?

THERMODYNAMICALLY STABLE SYSTEM/ 3

$$\frac{\partial^2 S_1(E^*)}{\partial E^2} + \frac{\partial^2 S_2(E_{total} - E^*)}{\partial E^2} < 0$$

$$\frac{\partial S}{\partial E} = \frac{1}{T}$$

$$\frac{\partial^2 S}{\partial E^2} = \frac{\partial}{\partial E} \left(\frac{1}{T} \right) = -\frac{1}{T^2} \frac{\partial T}{\partial E} = -\frac{1}{T^2 C}$$

We want the sum of the two $\frac{\partial^2 S}{\partial E^2}$ terms to be negative:

$$\longrightarrow C > 0$$

AN ASIDE: STIRLING'S FORMULA

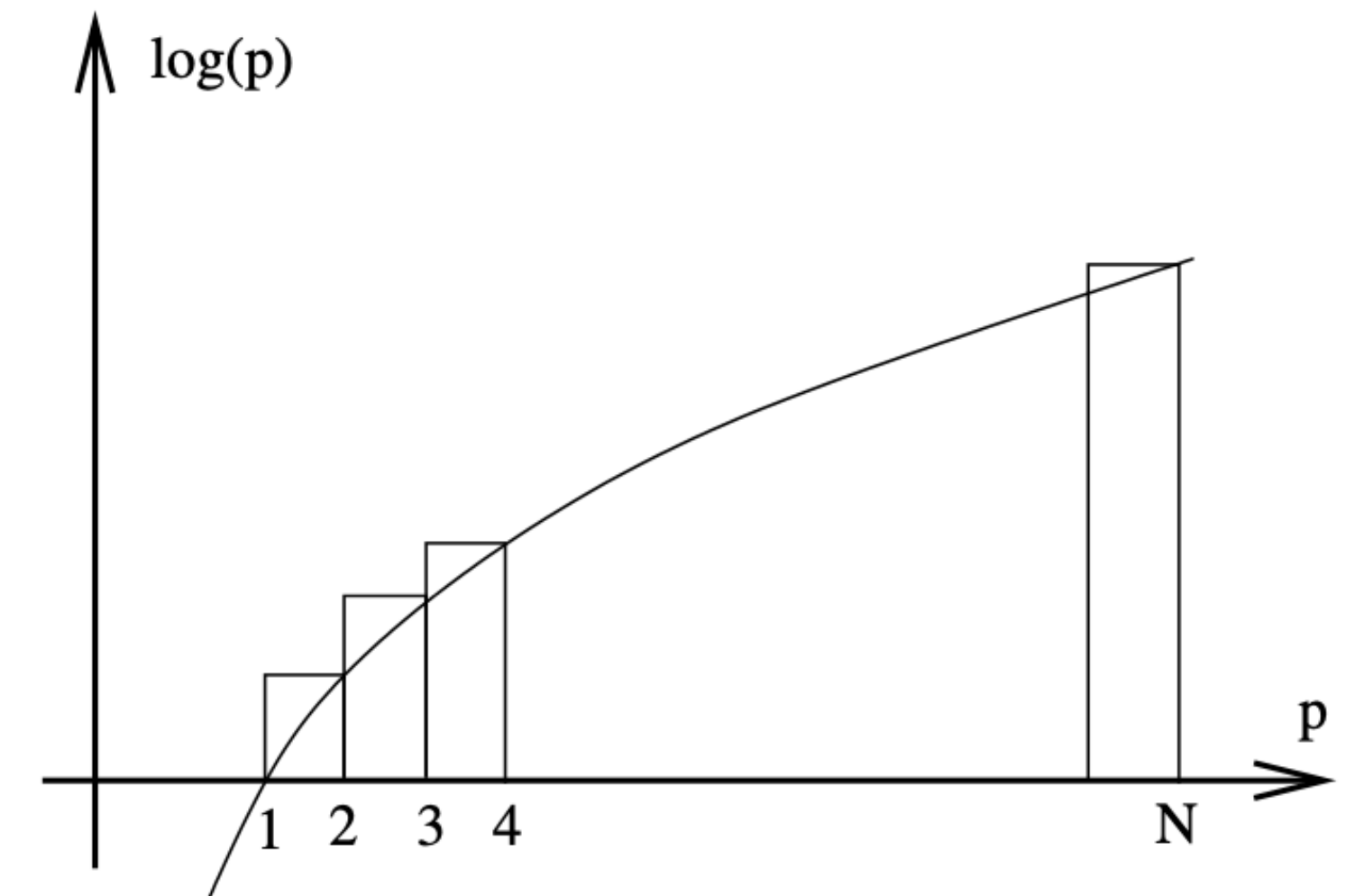
Approximation for $\log(N!) \sim N \log N - N$, with $N \gg 1$

Proof:

$$N! = 1 \times 2 \times \dots \times N$$

$$\log N! = \log 1 + \log 2 + \dots + \log N = \sum_{p=1}^N \log p \sim \int_0^N dp \log p$$

~same as 1 with large N



We check that $\int dp \log p = p \log p - p$ by checking $\frac{d}{dp} (p \log p - p) = \log p$

$$\frac{d}{dp} (p \log p - p) = p \frac{1}{p} + \log p - 1 = \log p$$

$$\log N! \sim p \log p - p \Big|_0^N = N \log N - N$$

THE TWO STATE SYSTEM/ 1

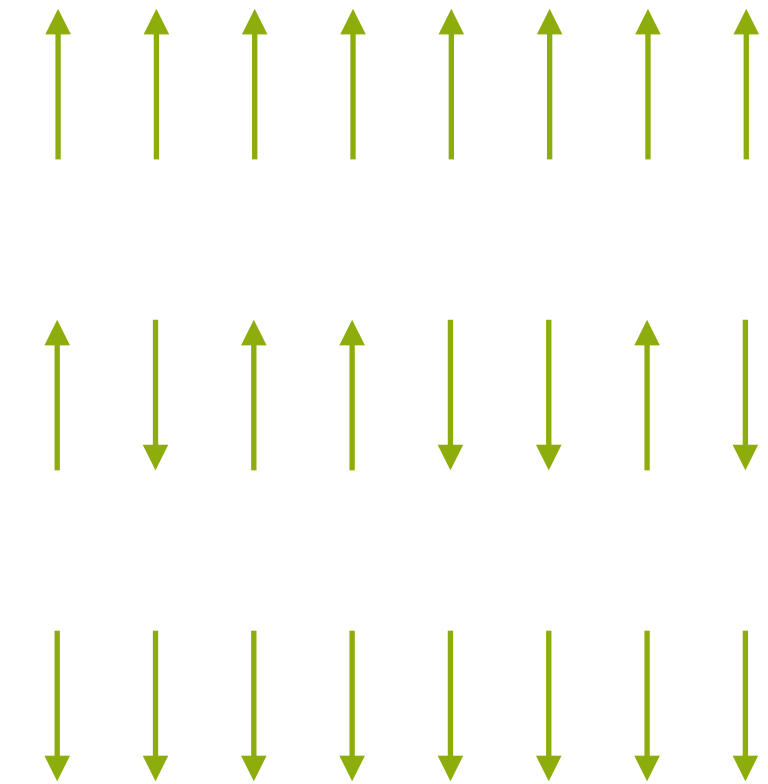
N **non-interacting** particles, either spin up $|\uparrow\rangle$ or spin down $|\downarrow\rangle$

Energy of the states (by assumption):

$$E_{\downarrow} = 0$$

$$E_{\uparrow} = \epsilon$$

You pay an energy cost of ϵ for each spin up



System with N spins up, $N_{\downarrow} = N - N_{\uparrow}$ spins down: $E = N_{\uparrow}\epsilon$

Number of states with energy $E = N_{\uparrow}\epsilon$: number of ways of picking $N_{\uparrow}$ particles out of a total of N :

Combinations, not permutations, because order does not matter

$$\Omega(E) = \frac{N!}{N_{\uparrow}!(N - N_{\uparrow})!}$$

Entropy:

$$S(E) = k_B \log \left(\frac{N!}{N_{\uparrow}!(N - N_{\uparrow})!} \right)$$

Stirling's formula

Combinations: order does not matter

$$C(n, k) = \frac{n!}{(n - k)!k!}$$

Permutations: order matters

$$P(n, k) = \frac{n!}{(n - k)!}$$

THE TWO STATE SYSTEM/ 2

$$\log(N!) \sim N \log N - N, \text{ with } N \gg 1$$

$$S(E) = k_B \log \left(\frac{N!}{N_{\uparrow}!(N - N_{\uparrow})!} \right)$$

Stirling

$$\begin{aligned} S(E) &= k_B [N \log N - N - N_{\uparrow} \log N_{\uparrow} + N_{\uparrow} - (N - N_{\uparrow}) \log(N - N_{\uparrow}) + (N - N_{\uparrow})] = \\ &+ N_{\uparrow} \log N - N_{\uparrow} \log N \\ &= -k_B \left[(N - N_{\uparrow}) \log \left(\frac{N - N_{\uparrow}}{N} \right) + N_{\uparrow} \log \left(\frac{N_{\uparrow}}{N} \right) \right] = \\ &= -k_B N \left[\left(1 - \frac{E}{N\epsilon} \right) \log \left(1 - \frac{E}{N\epsilon} \right) + \frac{E}{N\epsilon} \log \left(\frac{E}{N\epsilon} \right) \right] \end{aligned}$$

$E = N_{\uparrow}\epsilon$

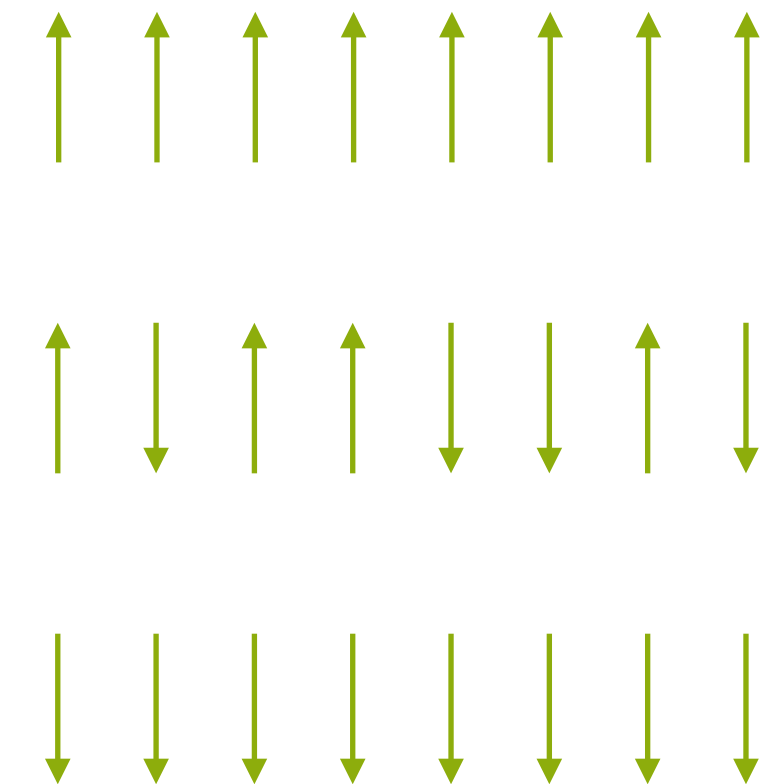
Plugging $E = 0, N\epsilon, N\epsilon/2$ in the eq. above:

$$S(0) = 0, S(N\epsilon) = 0, S(N\epsilon/2) = Nk_B \log 2$$

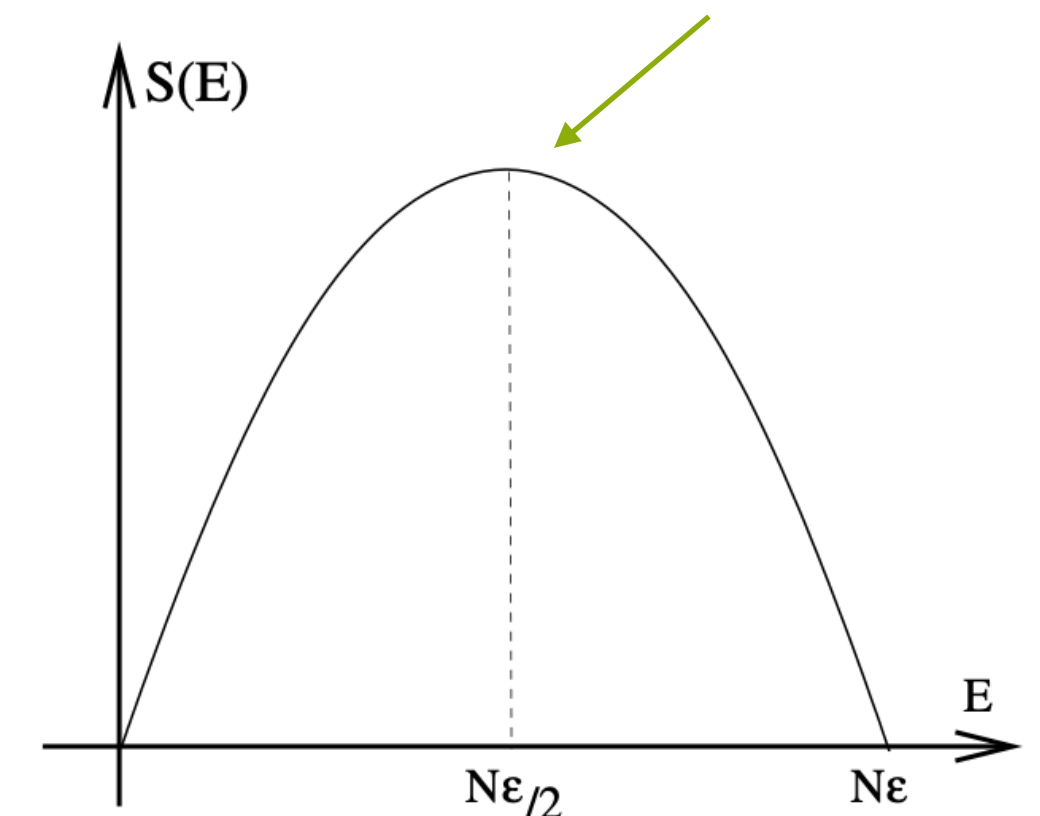
Q: can you explain this, in terms of available microstates?

T of system with energy E:

$$\frac{1}{T} = \frac{\partial S}{\partial E} = \frac{k_B}{\epsilon} \log \left(\frac{N\epsilon}{E} - 1 \right)$$



$$S(N\epsilon/2) = Nk_B \log 2$$



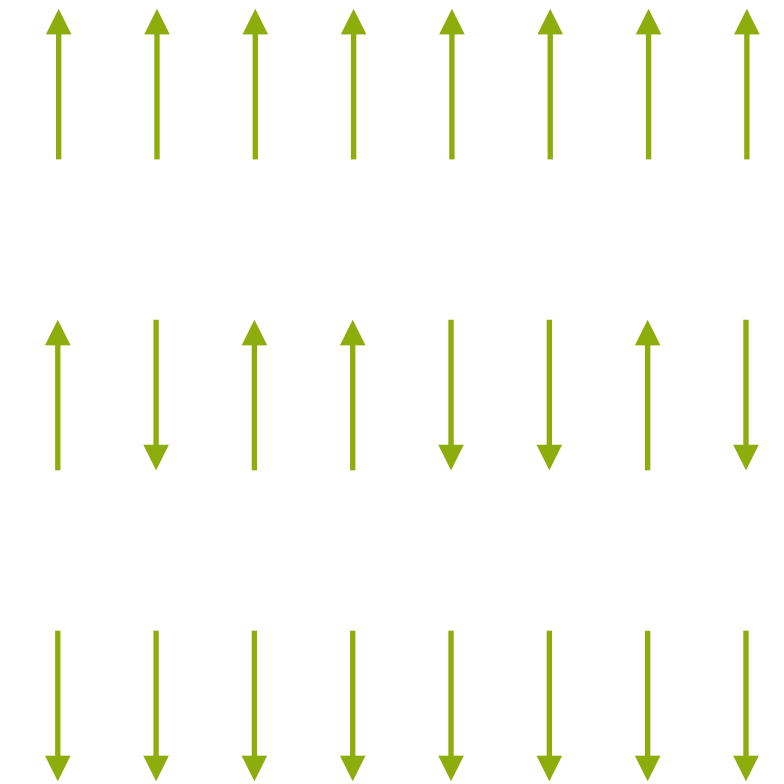
THE TWO STATE SYSTEM/ 3

Taking the exp:

$$\boxed{\frac{E}{N\epsilon}} = \frac{1}{e^{\epsilon/k_B T} + 1} = \boxed{\frac{N_{\uparrow}}{N}}$$

$$E = N_{\uparrow}\epsilon$$

Fraction of particles with spin up, with temperature T



Q: what happens with $T \rightarrow \infty$?

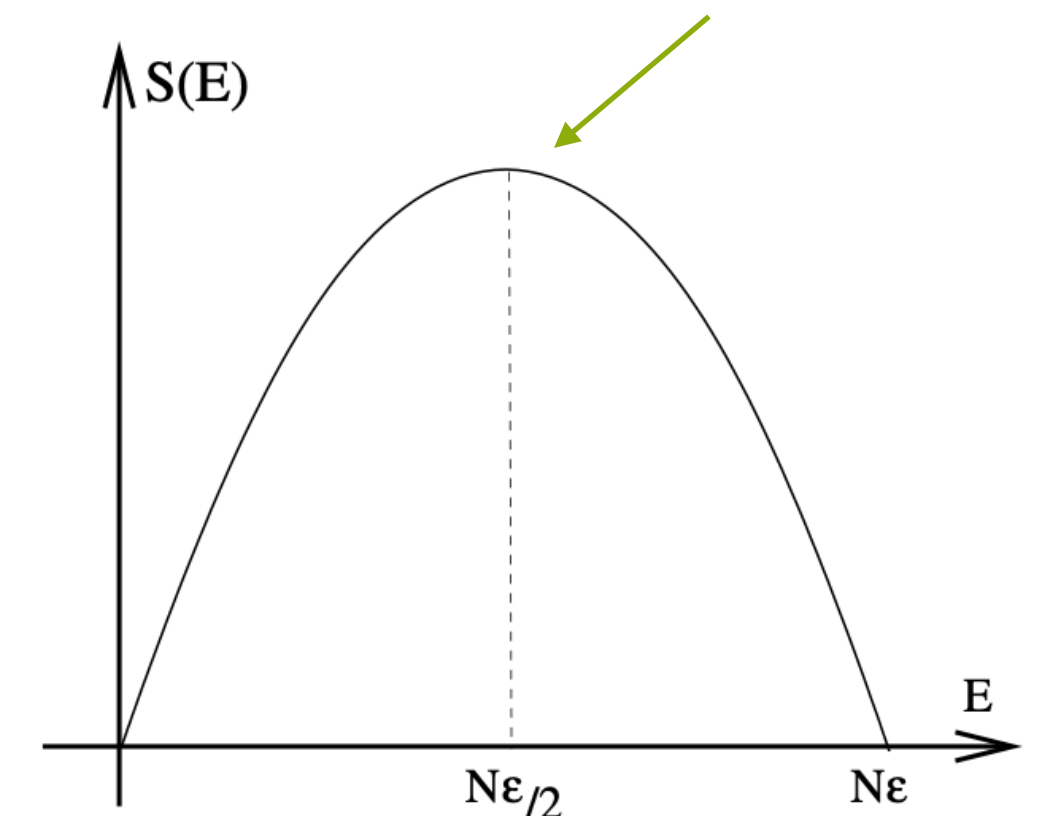
(see later)

Q: what happens with the temperature, with $E > N\epsilon/2$? $\frac{1}{T} = \frac{\partial S}{\partial E}$

Systems with “negative temperature:” adding energy, entropy decreases, which means the number of microstates decreases

In the lab, one can switch all spins in a system, for a short time

$$S(N\epsilon/2) = Nk_B \log 2$$



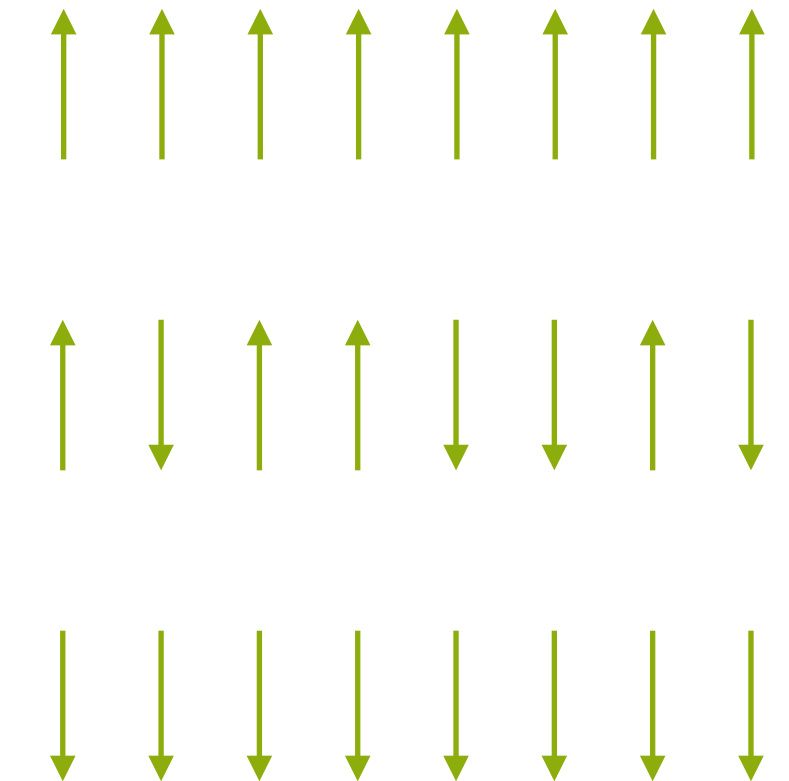
THE TWO STATE SYSTEM - $T \rightarrow \infty$

Taking the exp:

$$\frac{E}{N\epsilon} = \frac{1}{e^{\epsilon/k_B T} + 1} = \frac{N_{\uparrow}}{N}$$

$E = N_{\uparrow}\epsilon$

Fractions of particles with spin up, with temperature T



Q: what happens with $T \rightarrow \infty$?

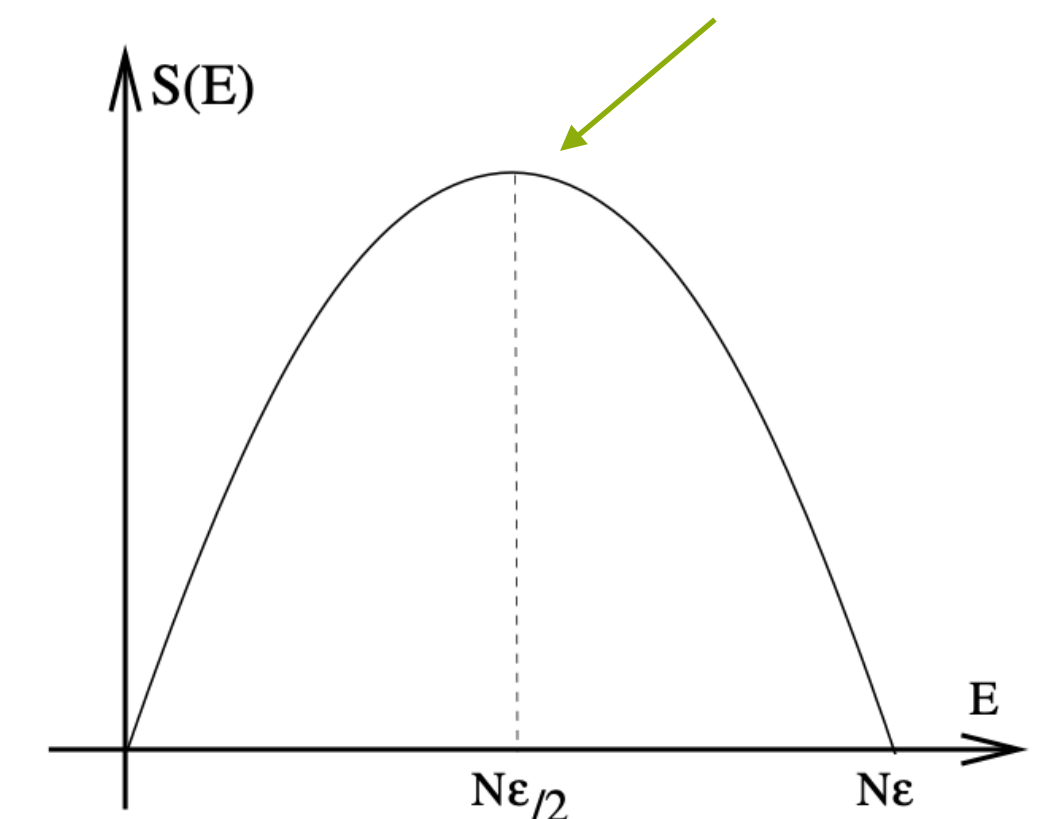
$$\frac{E}{N\epsilon} = \frac{1}{2} = \frac{N_{\uparrow}}{N}$$

$$\frac{N_{\uparrow}}{N} = 1/2; \text{ half spins up, half done}$$

$$E = N\epsilon/2$$

We are at the peak of the entropy

$$S(N\epsilon/2) = Nk_B \log 2$$



THE TWO STATE SYSTEM: HEAT CAPACITY AND THE SCHOTTKY ANOMALY

Energy of the two state system:

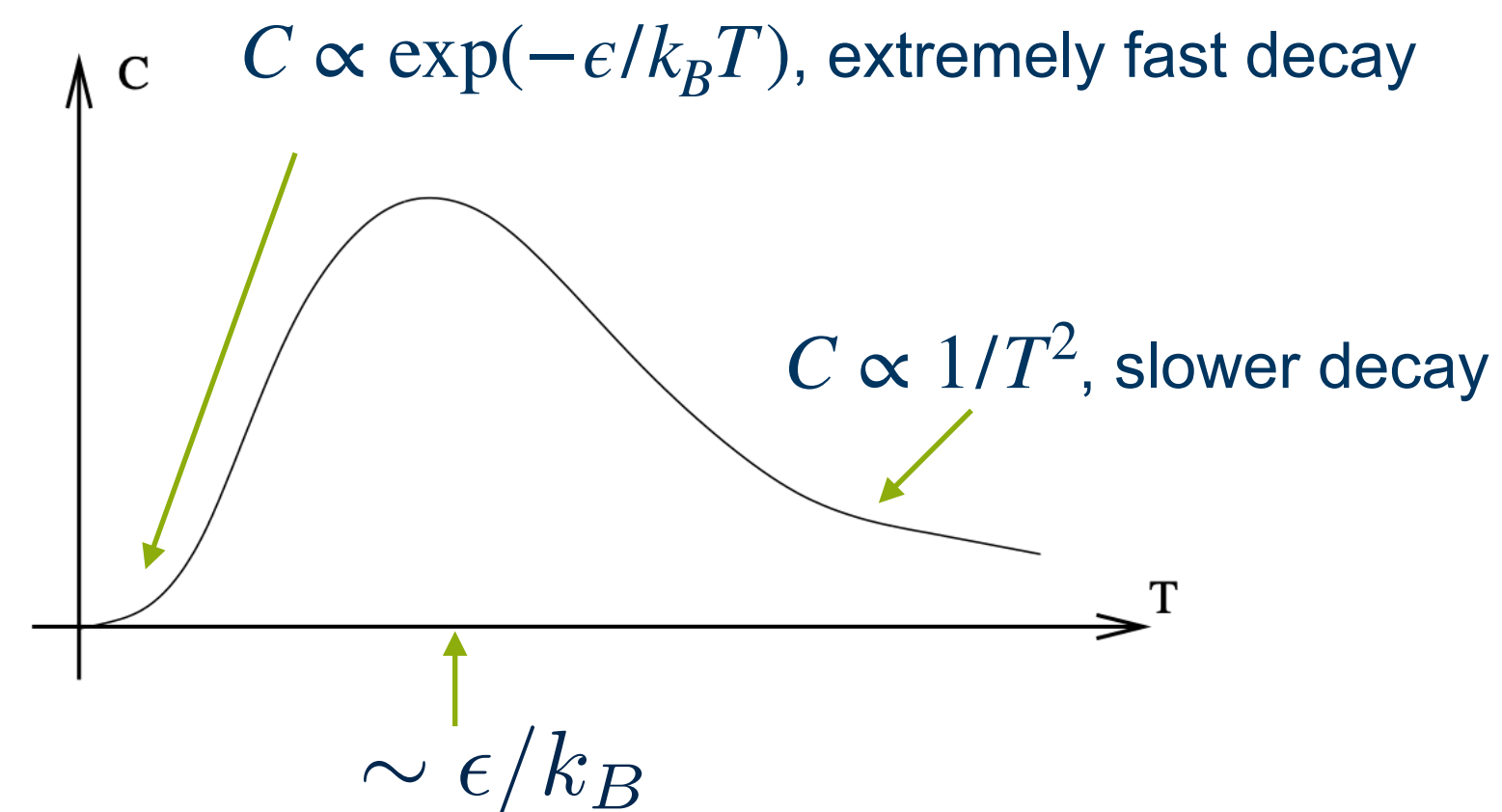
$$\frac{E}{N\epsilon} = \frac{1}{e^{\epsilon/k_B T} + 1} = \frac{N_{\uparrow}}{N}$$

Heat capacity of the two state system:

$$C = \frac{dE}{dT} = \frac{N\epsilon^2}{k_B T^2} \frac{e^{\epsilon/k_B T}}{(e^{\epsilon/k_B T} + 1)^2}$$

$C \propto N$, common property

From functional analysis, we calculate $C(T \rightarrow 0)$, $C(T \rightarrow \infty)$, $\max(C(T))$



Contribution to the heat capacity **from spins**

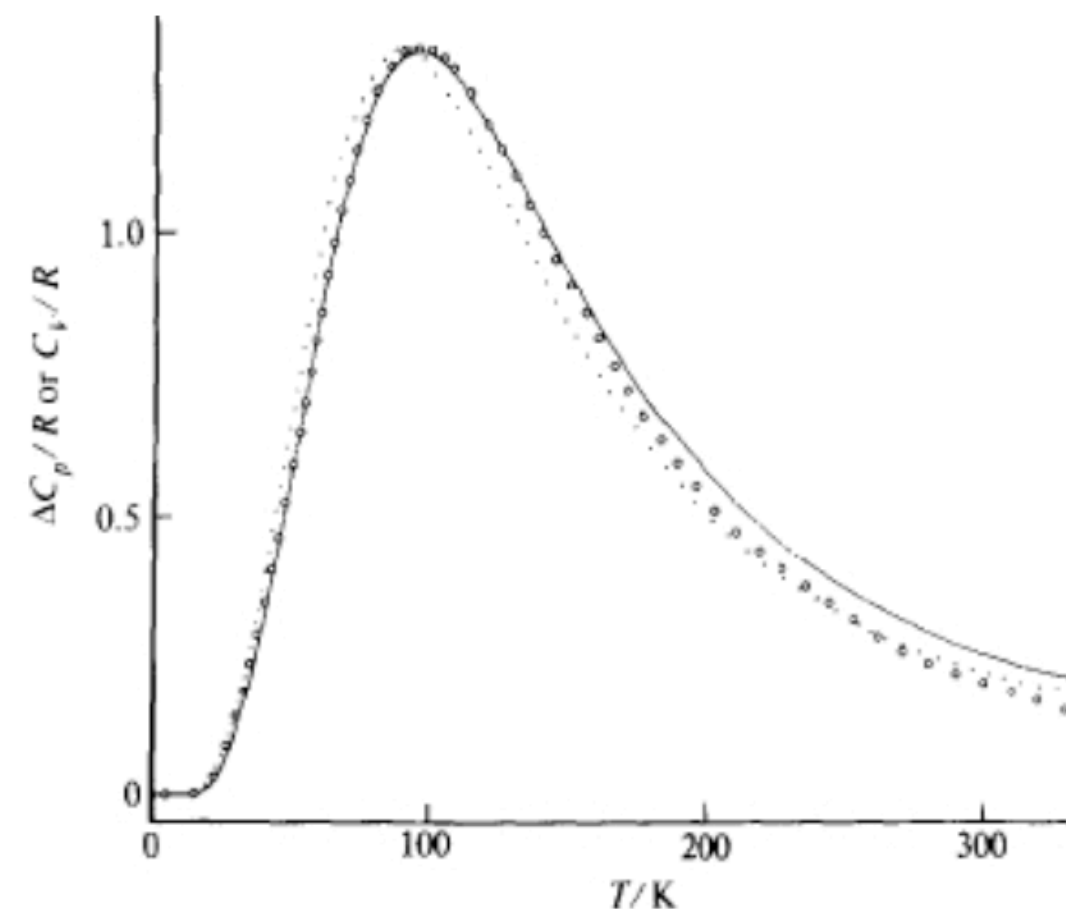
← This is not what C usually looks like, since the spin contribution in most materials is small with respect to phonon contributions or conducting electrons

C from phonons/ conducting electrons: C decreases monotonically with T , no 'bump'

THE SCHOTTKY ANOMALY

Schottky anomaly: in certain materials (e.g. paramagnetic salts), the heat capacity presents a ‘bump’ at $T \rightarrow 0$.

This is the spin contribution to heat capacity, which is not completely dwarfed by phonon contribution ($C \rightarrow 0$ monotonically when $T \rightarrow 0$)



Chirico and Westrum, 1980: spin contribution to the heat capacity of $\text{Tb}(\text{OH})_3$, Terbium (III) hydroxide

PRESSURE AND VOLUME/ 1

In the spin problem, particles were fixed. Now: what happens if particles can move? → **Volume** V of the system

New(-ish) definition of entropy:

$$S(E, V) = k_B \log \Omega(E, V)$$

New(-ish) definition of temperature:

$$\frac{1}{T} = \left. \frac{\partial S}{\partial E} \right|_V \quad \leftarrow \text{Keep } V \text{ fixed when differentiating}$$

Definition of pressure (i.e.: what happens when we differentiate S w.r.t. V):

$$p = T \frac{\partial S}{\partial V}$$

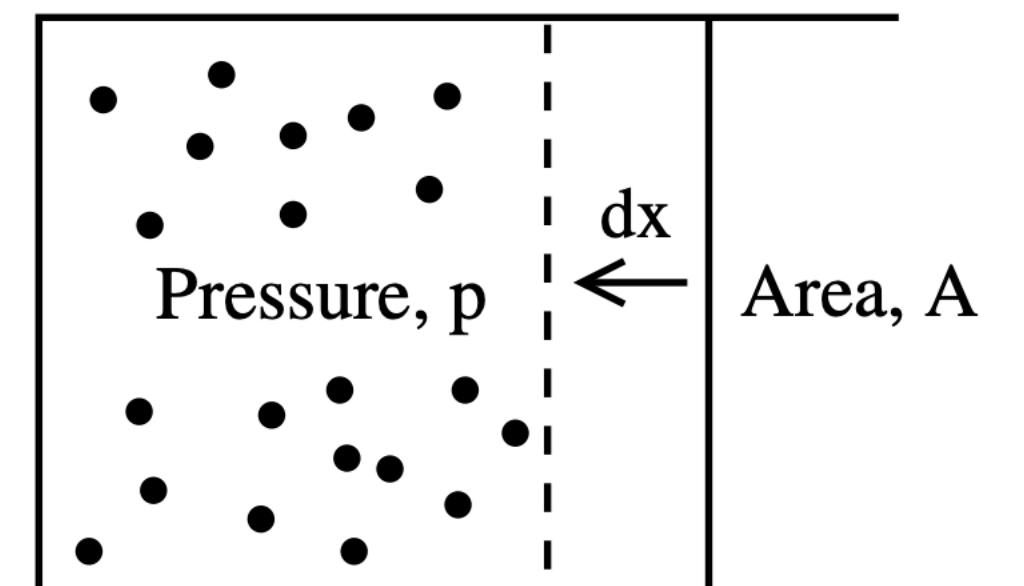


Movable partition: partition does not move if $\frac{\partial S_1}{\partial V_1} = \frac{\partial S_2}{\partial V_2}$, or $p_1 = p_2$

Now S depends on E and V :

$$dS = \underbrace{\frac{\partial S}{\partial E}}_{\frac{1}{T}} dE + \underbrace{\frac{\partial S}{\partial V}}_{\frac{p}{T}} dV$$

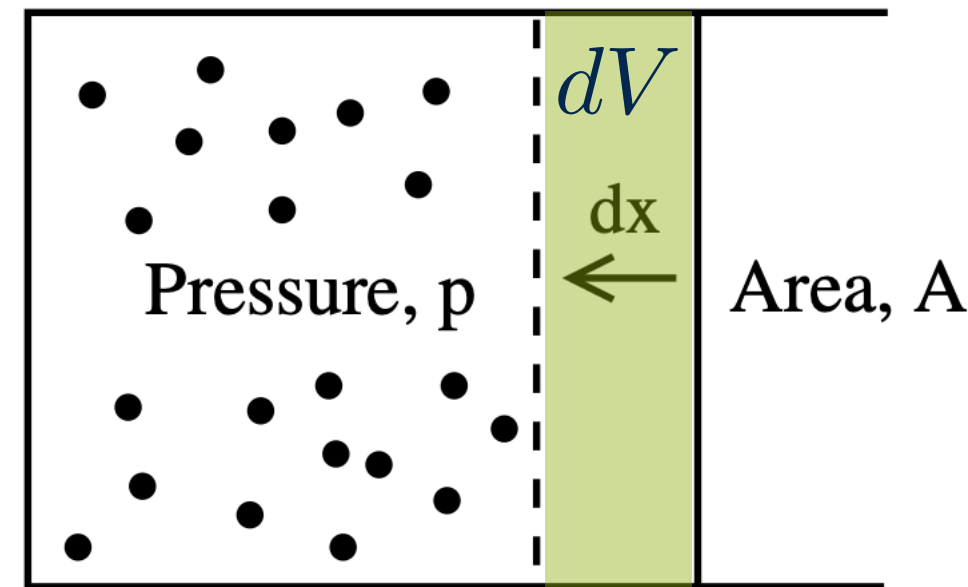
$$dE = TdS - pdV$$



FIRST LAW OF THERMODYNAMICS

(1) (2)

$$dE = TdS - pdV$$



What does (2) mean?

$$pdV = \frac{F}{A} A dx = F dx = dW$$

pressure= force/ unit surface

dVolume= area x displacement

dWork= force x displacement

The sign: squeezing the volume, $dV < 0$, the energy of the system increases, $dE > 0 \rightarrow$ you need the - sign

(2): variation of energy due to **work** done on the system

(1): variation of energy due to **heat** absorbed from the surrounding (internal energy)

First law of thermodynamics: conservation of energy for a system at finite temperature

$$dE = TdS - pdV$$

“Energy is conserved”

BACK TO HEAT CAPACITY

Before:

$$C = \frac{\partial E}{\partial T}$$

Now: heat capacity *at fixed volume*

$$C_V = \left. \frac{\partial E}{\partial T} \right|_V$$

I law with $dV=0$:

$$dE = TdS - \cancel{pdV}$$

Here we find the “heat”, all good

$$C_V = \left. \frac{\partial E}{\partial T} \right|_V = \left. \frac{TdS}{\partial T} \right|_V = T \left. \frac{\partial S}{\partial T} \right|_V$$

Also: heat capacity *at fixed pressure*

$$C_p = T \left. \frac{\partial S}{\partial T} \right|_p$$

MICROCANONICAL VS CANONICAL ENSEMBLE

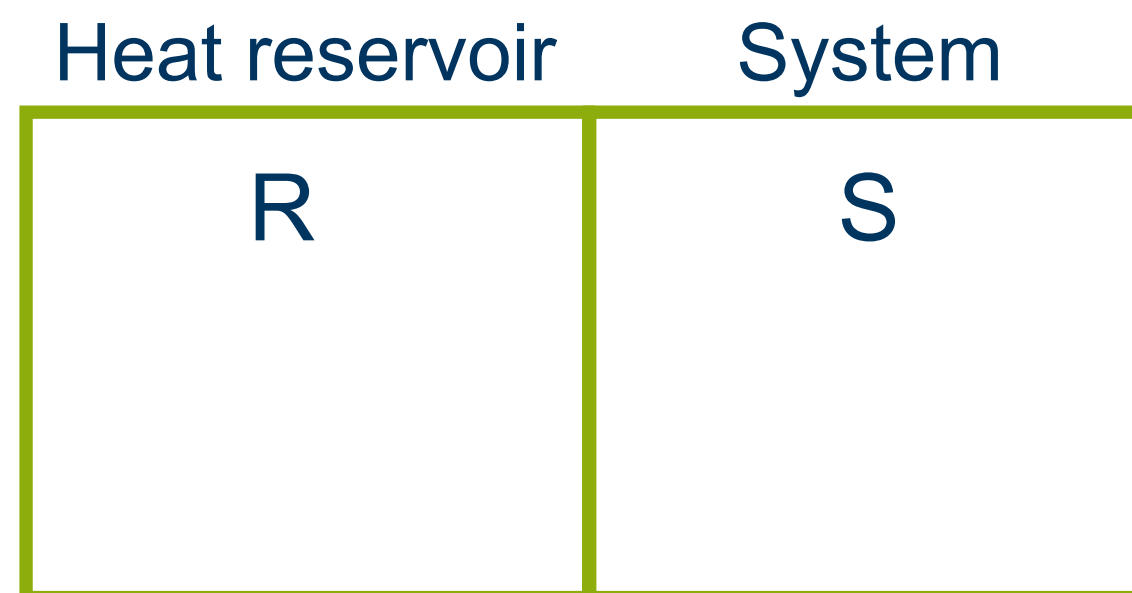
Microcanonical ensemble:

- The system is at fixed E , all accessible microstates have the same energy E
- All accessible microstates are equally likely
- Probability of given state $|n\rangle$

Q: what does 'accessible microstate' mean? What is the probability distribution in the **microcanonical** environment?

Now, a more realistic scenario: our system S is in contact and in equilibrium with an heat reservoir R at temperature T , with $E_R \gg E_S$

Q: can you think of some example?



S can absorb energy or donate energy to R , without E_R or T changing (significantly)

→ **S is at fixed temperature T** , rather than at fixed energy E

→ **canonical ensemble**

$$E_R \gg E_S$$

$$T_R = T_S = T$$

PROBABILITY DISTRIBUTION IN CANONICAL ENSEMBLES/ 1

As before, we look for the probability distribution of the system

Number of microstates of the **combined system**, S + R, with $|n\rangle$ the microstates in S with energy E_n :

$$E_R = E_{total} - E_n \quad E_n: \text{energy of the state } |n\rangle \text{ of S}$$

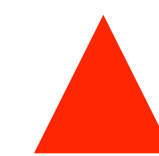
$$\Omega(E_{total}) = \sum_n \Omega_R(E_{total} - E_n) := \sum_n \exp\left(\frac{S_R(E_{total} - E_n)}{k_B}\right)$$

This is the entropy of the **reservoir**

Sum over the states $|n\rangle$ with energy $E = E_n$ of S, not over the energy levels (we'll talk about degeneracy later)

Q: did I forget something here? What did we say about available microstates in a combined system?

Definition of entropy: $S = k_B \log \Omega$



Careful: "fast" notation. What is a more accurate notation?

$E_n \ll E_{total} \rightarrow$ Taylor expansion of S_R :

$$S_R(E_{total} - E_n) \sim S_R(E_{total}) - E_n \frac{\partial S_R}{\partial E_{total}}$$

Number of microstates in combined system, S + R:

$$\Omega(E_{total}) \sim \sum_n \exp\left(\frac{S_R(E_{total})}{k_B} - \frac{\partial S_R}{\partial E_{total}} \frac{E_n}{k_B}\right)$$

$$\frac{\partial S_R}{\partial E_{total}} = \frac{1}{T}$$

Q; why T and not T_{total} ?

PROBABILITY DISTRIBUTION IN CANONICAL ENSEMBLES/ 2

Number of microstates in combined system, S + R:

$$\Omega(E_{total}) \sim \sum_n \exp \left(\frac{S_R(E_{total})}{k_B} - \frac{1}{T} \frac{E_n}{k_B} \right) = \sum_n \exp \left(\frac{S_R(E_{total})}{k_B} \right) \exp \left(-\frac{E_n}{k_B T} \right) =$$

Entropy/ energy of the reservoir *do not depend* on the state $|n\rangle$ of the system S (assumption from before) $\leftarrow$

$$= \exp \left(\frac{S_R(E_{total})}{k_B} \right) \sum_n \exp \left(-\frac{E_n}{k_B T} \right)$$

Usual assumption: **all accessible energy states of the S + R system are equally likely**

Number of states of the combined S+R system with S in $|n\rangle$: $\exp \left(\frac{S_R(E_{total})}{k_B} \right) \exp \left(-\frac{E_n}{k_B T} \right)$

Probability of $|n\rangle$: ratio of the number of states in S+R with S in $|n\rangle$ / total number of states

$$p(n) = \frac{e^{S_R(E_{total})/k_B} e^{-E_n/k_B T}}{\sum_m e^{S_R(E_{total})/k_B} e^{-E_m/k_B T}} = \frac{e^{-E_n/k_B T}}{\sum_m e^{-E_m/k_B T}}$$

Sum on all possible states of S

PROBABILITY DISTRIBUTION IN CANONICAL ENSEMBLES/ 3

Probability of microstate $|n\rangle$ in the canonical ensemble:

$$p(n) = \frac{e^{-E_n/k_B T}}{\sum_m e^{-E_m/k_B T}} \leftarrow \text{Boltzmann distribution}$$

- The probability distribution *does not* depend on the characteristics of the reservoir, only on the characteristics of the system S and on the equilibrium temperature T

• The denominator is a normalization factor, so that $\sum_n p(n) = 1$

- Dependence of probability of $|n\rangle$ on the associated energy E_n

Q: what is more probable in the canonical ensemble? $E_n \ll k_B T$ or $E_n \gg k_B T$?

Q: does the ground state (== state with lower energy) become more or less probable with decreasing temperature?

PARTITION FUNCTION

Probability of microstate $|n\rangle$ in the macrocanonical ensemble:

$$p(n) = \frac{e^{-E_n/k_B T}}{\sum_m e^{-E_m/k_B T}}$$

$$\beta := \frac{1}{k_B T}$$

Normalization factor in the Boltzmann probability distribution: $Z = \sum_m e^{-\beta E_m}$

Partition function: the partition function (or better, the logarithm of the partition function) contains the most important info on the state of the system!

Q: what is the partition function of system composed of two **independent** systems, S_1 and S_2 , with energy $E_n^{(1)}$ and $E_m^{(2)}$ respectively?

PARTITION FUNCTION OF INDEPENDENT SYSTEMS

Partition function of a single system:

$$Z = \sum_m e^{-\beta E_m}$$

Partition function of two independent systems, with energy $E_n^{(1)}$ and $E_m^{(2)}$

Total energy: $E_{tot} = E_n^{(1)} + E_m^{(2)}$

$$Z = \sum_{n,m} e^{-\beta(E_n^{(1)} + E_m^{(2)})} = \sum_{n,m} e^{-\beta E_n^{(1)}} e^{-\beta E_m^{(2)}} = \sum_n e^{-\beta E_n^{(1)}} \sum_m e^{-\beta E_m^{(2)}} = Z_1 Z_2$$

For *independent* systems, partition functions *multiply*

PARTITION FUNCTION AND AVERAGE ENERGY

Some calculations:

$$\frac{\partial}{\partial \beta} \log Z = \frac{1}{Z} \frac{\partial Z}{\partial \beta} = \sum_n \frac{1}{Z} (-E_n) e^{-\beta E_n}$$

Average energy of the system:

$$\langle E \rangle = \sum_n p(n) E_n = \sum_n \frac{E_n e^{-\beta E_n}}{Z}$$

Partition function:

$$Z = \sum_n e^{-\beta E_n}$$

Average energy as a function of the partition function:

$$\langle E \rangle = -\frac{\partial}{\partial \beta} \log Z$$

← If we know the partition function of a system, we can derive its average energy

SUMMARY

- $\frac{1}{T} = \frac{\partial S}{\partial E}$; with this definition, energy flows in the direction we expect
- $C = \frac{\partial E}{\partial T}$; heat capacity is a link between the micro (S) and the macro-world; allows us to measure S
- Two-state system: the Schottky anomaly tells us that the approach works
- First law of thermodynamics: $dE = TdS - pdV$

References: Tong, Sec. 1.2.2 to 1.2.5